

FSA-v5 Technical Implementation Guide

A Reference for Engineers Porting the Model into `smc2fc`

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Abstract

This document is the definitive technical reference for the FSA-v5 physiological model as implemented in `models/fsa_high_res/`. Its audience is the research engineer who must integrate the model into the `smc2fc` inference and control framework. It covers the full mathematical specification (latent SDE + observation likelihood), the v5 deconditioning extension that produces the closed-island basin topology, the chance-constrained probabilistic cost functional, the numerical integration protocol, the parameter freeze policy and its Fisher-information justification, end-to-end test scenarios on synthetic data, and a list of bugs and gotchas surfaced during the v5 implementation pass. A complete parameter reference is included as an appendix.

This document deliberately omits the historical narrative arcs of v2, v3, and v4 that appear in the longer modelling notes (`main.tex`). For background motivation, that document is the reference; for implementation, this document is the reference.

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1 Overview

1.1 What this guide covers

The FSA-v5 model is a 6-dimensional stochastic differential equation (SDE) with five observation channels. Forward simulation, Bayesian inference, and closed-loop control of this model are all supported by the code in `models/fsa_high_res/`. This guide explains:

- The mathematical specification — every latent equation, every observation channel, every parameter.
- The closed-island basin topology of v5 — the model’s prediction that there are three physiologically meaningful steady states (mono-stable healthy, sedentary collapse, over-training collapse).
- The chance-constrained probabilistic cost functional that defines what “optimal training” means under v5.
- The numerical integration protocol (15-minute bin grid, no sub-stepping, state-clipping) and its identifiability consequences.
- Which parameters are estimated and which are frozen, and the Fisher-information argument behind that choice.
- Three constant- Φ deterministic test scenarios that an engineer can run during the port to verify the implementation works as expected.
- Common bugs and gotchas surfaced during the v5 implementation pass — read this section before you start porting.

1.2 What this guide deliberately omits

This guide does not cover the historical evolution from v2 through v4, the FSA-v3 bimodal extension, the v4 Variable-Dose extension, or the gradient-OT control formulation. Those are in `main.tex` for background. This document treats v5 as the canonical model.

1.3 Files and the integration boundary

The complete v5 implementation lives in `models/fsa_high_res/`. The integration boundary — everything `smc2fc` should import from — is the package’s `__init__.py`, which re-exports the following symbols:

Symbol	Source file	Role
<code>HIGH_RES_FSA_V5_MODEL</code>	<code>simulation.py</code>	Forward-sim <code>SDEModel</code>
<code>HIGH_RES_FSA_V5_ESTIMATION</code>	<code>estimation.py</code>	Inference <code>EstimationModel</code>
<code>StepwisePlant</code>	<code>_plant.py</code>	Closed-loop synthetic plant
<code>evaluate_chance_constrained_cost</code>	<code>control_v5.py</code>	v5 cost (eqs. 21, 22)
<code>find_A_sep_v5</code>	<code>control_v5.py</code>	Separatrix root-finder
<code>TRUTH_PARAMS_V5</code>	<code>_dynamics.py</code>	v5 truth parameter dict
<code>DEFAULT_PARAMS_V5</code>	<code>simulation.py</code>	v5 truth + observation coefficients
<code>DEFAULT_INIT</code>	<code>simulation.py</code>	6D initial state defaults
<code>drift_jax</code>	<code>_dynamics.py</code>	Canonical drift (single source of truth)
<code>diffusion_state_dep</code>	<code>_dynamics.py</code>	State-dependent diffusion
<code>BINS_PER_DAY, DT_BIN_DAYS</code>	<code>simulation.py</code>	Time-grid constants
<code>circadian, circadian_jax</code>	<code>simulation.py</code>	Cosine circadian regressor
<code>build_control_spec_v5</code>	<code>control.py</code>	Gradient-OT spec (legacy/comparison)

A v4 alias (`HIGH_RES_FSA_V4_*`) is kept for back-compat with existing tests; it is numerically identical to v5 with the Hill deconditioning amplitudes pinned at zero.

1.4 How to read this guide

Sections 2–3 are the model specification: read them top to bottom in order. Sections 4–5 give the v5 contributions (basin geometry + probabilistic cost) that distinguish v5 from prior versions. Sections 6–7 are operational — how to integrate and what to estimate. Sections 8–9 are practical — test scenarios and bugs to anticipate. The appendix is a parameter reference. Code snippets throughout cite specific files in the package.

2 Latent dynamics: the FSA-v5 SDE

2.1 State vector

The latent state at time t is the 6-vector

$$y(t) = (B(t), S(t), F(t), A(t), K_{FB}(t), K_{FS}(t))^T, \quad (1)$$

with the components defined as follows.

Symbol	Range	Physiological role	Diffusion family
B	$[0, 1]$	Aerobic fitness (Banister chronic capacity)	Jacobi, $\sqrt{B(1-B)}$
S	$[0, 1]$	Strength capacity (Banister chronic capacity)	Jacobi, $\sqrt{S(1-S)}$
F	$[0, \infty)$	Unified fatigue pool	CIR, $\sqrt{F}$
A	$[0, \infty)$	Autonomic amplitude (HRV / autonomic resilience)	CIR, $\sqrt{A + \varepsilon}$
K_{FB}	$[0, \infty)$	Aerobic fatigue gain (Busso variable-dose)	CIR, $\sqrt{K_{FB}}$
K_{FS}	$[0, \infty)$	Strength fatigue gain (Busso variable-dose)	CIR, $\sqrt{K_{FS}}$

Two operating-point reference constants are baked into the G1-reparametrised drift: $A_{\text{TYP}} = 0.10$ (typical autonomic amplitude) and $F_{\text{TYP}} = 0.20$ (typical fatigue). They calibrate the model around a physiologically representative state.

2.2 Drift equations

The deterministic part of the SDE is implemented in `models/fsa_high_res/_dynamics.py:drift_jax` as the canonical single source of truth. The six equations are:

$$\dot{B} = \kappa_B a_B(A) \Phi_B - \frac{B}{\tau_B}, \quad a_B(A) = \frac{1 + \epsilon_{AB} A}{1 + \epsilon_{AB} A_{\text{TYP}}}, \quad (2)$$

$$\dot{S} = \kappa_S a_S(A) \Phi_S - \frac{S}{\tau_S}, \quad a_S(A) = \frac{1 + \epsilon_{AS} A}{1 + \epsilon_{AS} A_{\text{TYP}}}, \quad (3)$$

$$\dot{F} = K_{FB} \Phi_B + K_{FS} \Phi_S - \frac{a_F(A)}{\tau_F} F, \quad a_F(A) = \frac{1 + \lambda_A A}{1 + \lambda_A A_{\text{TYP}}}, \quad (4)$$

$$\dot{A} = \mu(B, S, F) A - \eta A^3, \quad (5)$$

$$\dot{K}_{FB} = \frac{K_{FB}^0 - K_{FB}}{\tau_K} + \mu_K \Phi_B, \quad \dot{K}_{FS} = \frac{K_{FS}^0 - K_{FS}}{\tau_K} + \mu_K \Phi_S. \quad (6)$$

The exogenous control input is the bimodal stimulus rate $\Phi(t) = (\Phi_B(t), \Phi_S(t))^T$ — aerobic and strength training loads respectively, both dimensionless.

Stuart–Landau bifurcation parameter (the v5 form). The autonomic equation (5) contains the multiplicative parameter $\mu(B, S, F)$, which is the locus of the v5 deconditioning extension. Its full v5 form is:

$$\mu(B, S, F) = \mu_0 + \mu_B B + \mu_S S - \mu_F F - \mu_{FF}(F - F_{\text{TYP}})^2 - \mu_{B-} h_n(B; B_{\text{dec}}) - \mu_{S-} h_n(S; S_{\text{dec}}), \quad (7)$$

where the v5 Hill-deconditioning weight is

$$h_n(x; x_{\text{dec}}) = \frac{x_{\text{dec}}^n}{x^n + x_{\text{dec}}^n} \in [0, 1]. \quad (8)$$

Setting $\mu_{B-} = \mu_{S-} = 0$ recovers the FSA-v4 form exactly, which is why $\mu_{\text{dec}*} = 0$ are the v4-recovering defaults in `TRUTH_PARAMS`; `TRUTH_PARAMS_V5` sets them to non-zero values that activate the closed-island topology of Section 4.

2.3 What each parameter controls — physiological intuition

Symbol	Meaning
τ_B, τ_S	Exponential decay time-scales of aerobic and strength capacities under zero stimulus. $\tau_B = 42\text{d}$, $\tau_S = 60\text{d}$ — so strength is slower to detrain than aerobic fitness.
κ_B, κ_S	Per-unit-stimulus accrual rates for B and S , respectively.
$\epsilon_{AB}, \epsilon_{AS}$	Autonomic boost: when A is high, training builds capacity faster ($a_*(A) > 1$).
τ_F	Fatigue clearance time-scale ($\sim 7\text{d}$ after autonomic correction).
λ_A	Autonomic acceleration of fatigue clearance: high A clears fatigue faster.
K_{FB}^0, K_{FS}^0	Baseline fatigue gains (untrained athlete). K_{Fi} rises above baseline under sustained training.
τ_K	Recovery time-scale of the fatigue sensitivity ($\sim 21\text{d}$, three weeks).
μ_K	“Damage” rate: how fast training pushes K above baseline.
μ_0	Baseline autonomic drive at zero load.
μ_B, μ_S	Coupling of fitness and strength to autonomic drive.
μ_F	Fatigue’s linear suppression of autonomic drive.
μ_{FF}	Quadratic penalty around F_{TYP} — captures the cliff where overtraining suppresses HRV.
η	Cubic damping of A — prevents runaway autonomic amplitude.
$B_{\text{dec}}, S_{\text{dec}}$	Chronic-fitness thresholds below which deconditioning kicks in.
μ_{B-}, μ_{S-}	Maximum autonomic-drive penalty when B (resp. S) is far below threshold.
n_{dec}	Hill exponent — steepness of the deconditioning transition (default 4).

2.4 Diffusion: state-dependent diagonal

The stochastic part of the SDE is diagonal with state-dependent scales, implemented in `_dynamics.py:diffusion_state_dep`. Each component has a physically-motivated form:

$$\sigma(y) = \text{diag}\left(\sigma_B \sqrt{B(1-B)}, \sigma_S \sqrt{S(1-S)}, \sigma_F \sqrt{F}, \sigma_A \sqrt{A + \epsilon_A}, \sigma_K \sqrt{K_{FB}}, \sigma_K \sqrt{K_{FS}}\right). \quad (9)$$

The Jacobi-form $\sqrt{x(1-x)}$ scales for B, S keep them strictly in $[0, 1]$; the CIR-form $\sqrt{x}$ scales for F, A, K_* keep them strictly non-negative. The five σ_* scalars and the floor ϵ_A are *frozen* in production — diffusion-side identifiability is weak and standard practice is not to estimate diffusion scales jointly with drift parameters.

2.5 Cascade structure

The drift has a feed-forward cascade structure under *constant* Φ : $K \rightarrow (B, S) \rightarrow F \rightarrow A$, with the only back-coupling being the autonomic state A entering the $a_*(A)$ multipliers. This

structure is what makes the v5 stability analysis tractable: at the slow-manifold equilibrium each forward block has a closed-form solution given A , and the entire 6D fixed-point reduces to a single scalar self-consistency equation in A (see Section 4).

3 Observation model

The observation likelihood is implemented in `models/fsa_high_res/estimation.py` (functions `_gaussian_obs_ll` and `obs_log_weight_fn`) and the synthetic-data samplers in `simulation.py` (`gen_obs_sleep`, `gen_obs_hr`, etc.).

3.1 Five observation channels

At each observation time t_k , with the latent state evaluated at that bin and a circadian regressor C_k , the model emits up to five observations:

$$\mu_k^{\text{HR}} = \text{HR}_{\text{base}} - \kappa_B^{\text{HR}} B_k + \alpha_A^{\text{HR}} A_k + \beta_C^{\text{HR}} C_k, \quad y_k^{\text{HR}} \sim \mathcal{N}(\mu_k^{\text{HR}}, \sigma_{\text{HR}}^2), \quad (10)$$

$$\mu_k^{\text{S}} = S_{\text{base}} + k_F F_k - k_{A,S} A_k + \beta_C^{\text{S}} C_k, \quad y_k^{\text{S}} \sim \mathcal{N}(\mu_k^{\text{S}}, \sigma_{\text{S}}^2), \quad (11)$$

$$\mu_k^{\text{st}} = \mu_{\text{step},0} + \beta_B^{\text{st}} B_k - \beta_F^{\text{st}} F_k + \beta_A^{\text{st}} A_k + \beta_C^{\text{st}} C_k, \quad y_k^{\text{st}} \sim \mathcal{N}(\mu_k^{\text{st}}, \sigma_{\text{st}}^2), \quad (12)$$

$$\mu_k^{\text{VL}} = \beta_S^{\text{VL}} S_k - \beta_F^{\text{VL}} F_k, \quad y_k^{\text{VL}} \sim \mathcal{N}(\mu_k^{\text{VL}}, \sigma_{\text{VL}}^2), \quad (13)$$

$$p_k^{\text{sleep}} = \sigma(k_C C_k + k_A A_k - \tilde{c}), \quad y_k^{\text{sleep}} \sim \text{Bernoulli}(p_k^{\text{sleep}}). \quad (14)$$

Linear-in-state. The four Gaussian channels are linear functions of the 4-dimensional sub-state (B, S, F, A) plus a circadian intercept; the K states are not directly observed. The Bernoulli sleep channel is a logistic in A and C . This makes the observation Jacobian H a constant-coefficient 4×6 matrix at each time, enabling the closed-form Kalman fusion in `estimation.py:propagate_fn_v5`.

Reading the channels physiologically.

- **HR** (heart rate, bpm): increases with autonomic amplitude, decreases with chronic aerobic fitness, modulated by circadian.
- **Stress** (subjective wellness score): increases with fatigue, decreases with autonomic amplitude.
- **Steps** (daily activity log): increases with both fitness and autonomic amplitude, decreases with fatigue.
- **Volume Load** (training session volume): observed only during training events — linear in strength capacity, with fatigue-driven reduction.
- **Sleep** (binary good/poor): logistic in autonomic amplitude and circadian.

3.2 Channel gating: sleep-aware sampling

Each Gaussian channel is emitted at the bin grid but is *gated* by the subject’s sleep / wake state. Production assumption: a canonical sleep window of 22:00–06:00 (8 hours), implementable from a single time-of-day flag. The gating policy is:

Channel	Gating	Rationale
HR	sleep-only	overnight resting HR is the most informative for B
Stress	wake-only	subjective scoring during waking hours
Steps	wake-only	step counter active only during wake
VolumeLoad	training sessions	one entry per training session ($\leq 1/\text{day}$)
Sleep label	every bin	sleep state itself is the obs

The dense overnight HR sampling is not optional: as Section 6 discusses, the production 15-minute bin grid combined with this sleep-gated HR is what makes the model identifiable. Replacing it with daily-summary HR observations inflates the FIM condition number by 20 orders of magnitude.

3.3 Circadian regressor

The circadian variable $C(t)$ is a single 24-hour cosine,

$$C(t) = \cos(2\pi t + \phi), \quad \phi := 0, \quad (15)$$

implemented in `simulation.py:circadian` (numpy) and `simulation.py:circadian_jax` (JAX). The phase ϕ is held at zero in production. The four β_C^* coefficients in equations (10)–(12) together with k_C in (14) describe the coupling of each channel to this shared regressor.

4 The three regimes: closed-island basin topology

The headline mathematical content of FSA-v5 is that, for constant training input $\Phi = (\Phi_B, \Phi_S)$, the long-time behaviour of the 6D system reduces to a single scalar self-consistency equation in A , and the basins of attraction in stimulus space have a closed-island geometry. This section gives the engineering-level summary; the full analysis is in `main.tex` §7 and §10.

4.1 Self-consistency equation for A

Under the cascade structure of Section 2.5, at constant Φ each forward block has a closed-form slow-manifold equilibrium parametrised by A :

$$K_{Fi}^*(\Phi_i) = K_{Fi}^0 + \tau_K \mu_K \Phi_i, \quad (16)$$

$$B^*(A; \Phi_B) = \tau_B \kappa_B a_B(A) \Phi_B, \quad (17)$$

$$S^*(A; \Phi_S) = \tau_S \kappa_S a_S(A) \Phi_S, \quad (18)$$

$$F^*(A; \Phi) = \tau_F (K_{FB}^* \Phi_B + K_{FS}^* \Phi_S) / a_F(A). \quad (19)$$

Substituting these into the v5 bifurcation parameter (eq. 7) defines the slow-manifold drive $\bar{\mu}(A; \Phi)$. The interior equilibria of A then solve

$$\bar{\mu}(A; \Phi) = \eta A^2, \quad (20)$$

together with the boundary equilibrium $A = 0$. Equation (20) has 0, 1, or 2 positive roots depending on Φ , giving rise to three qualitatively distinct regimes.

4.2 The three regimes

Mono-stable healthy. $\bar{\mu}(0; \Phi) > 0$: one positive root A^* . The boundary $A = 0$ is unstable, the interior A^* is stable. Healthy autonomic amplitude is the unique attractor regardless of initial condition.

Bistable. $\bar{\mu}(0; \Phi) < 0$ but the curve $g(A) = \bar{\mu}(A) - \eta A^2$ rises above zero somewhere: two positive roots $A_{\text{sep}} < A^*$. Both $A = 0$ and A^* are stable; A_{sep} is the unstable separatrix that partitions the basin. Initial $A_0 < A_{\text{sep}}$ collapses to zero, $A_0 > A_{\text{sep}}$ rises to A^* .

Mono-stable collapsed. g stays negative on $A > 0$: no positive roots. $A = 0$ is the global attractor. Autonomic amplitude is irrecoverable.

4.3 Closed-island geometry in (Φ_B, Φ_S) space

Figure 1 shows the central qualitative result. Under the v4 model (left) the healthy region is an open quarter-disc: sedentary ($\Phi = 0$) is healthy. Under the v5 modification (right) the boundary is a closed contour strictly interior to both axes: both $\Phi_B = 0$ (no aerobic) and $\Phi_S = 0$ (no strength) push outside the healthy region, so deconditioning by either route leads to collapse.

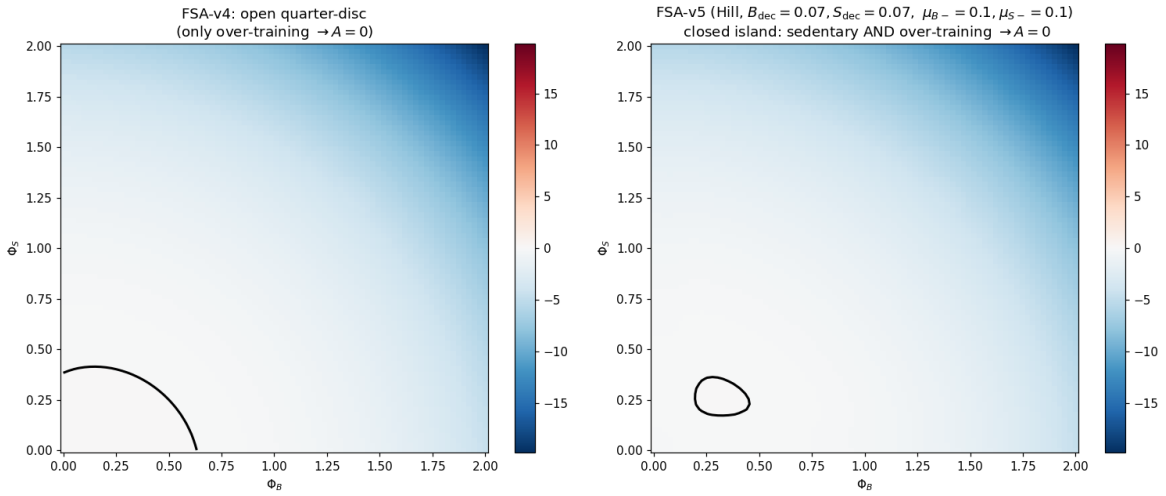


Figure 1: Transcritical bifurcation contour $\bar{\mu}(0; \Phi) = 0$ under FSA-v4 (left) and FSA-v5 (right). v4 admits a healthy regime down to $\Phi = 0$ and along both axes. The v5 Hill-deconditioning term ($B_{\text{dec}} = S_{\text{dec}} = 0.07$, $\mu_{B-} = \mu_{S-} = 0.10$, $n = 4$) closes the contour into a small island whose boundary is strictly interior to both coordinate axes. The medical “Goldilocks” / inverse-U structure emerges as a basin-topology change.

Figure 2 adds the saddle-node curve and the numerical regime classification. Inside the black transcritical contour: mono-stable healthy. Between black and magenta dashed: bistable annulus. Outside magenta: mono-stable collapsed.

4.4 Time-domain confirmation

The static analysis is confirmed by deterministic ODE rollouts under three constant- Φ scenarios from a common “trained athlete” initial state, shown in Figure 3. **Both** sedentary and over-training drive $A \rightarrow 0$ but via different mechanisms and on different time-scales:

- **Sedentary** ($\Phi = 0$): B and S decay exponentially at τ_B, τ_S . Once they cross $B_{\text{dec}}, S_{\text{dec}}$, the Hill terms in (7) saturate, $\bar{\mu}$ goes negative, and A collapses around day 100–150. Mediated by detraining of chronic capacities.
- **Moderate** ($\Phi = (0.30, 0.30)$, inside the v5 island): all states settle to a healthy steady state, $A^* \approx 0.55$. Goldilocks-band homeostasis.

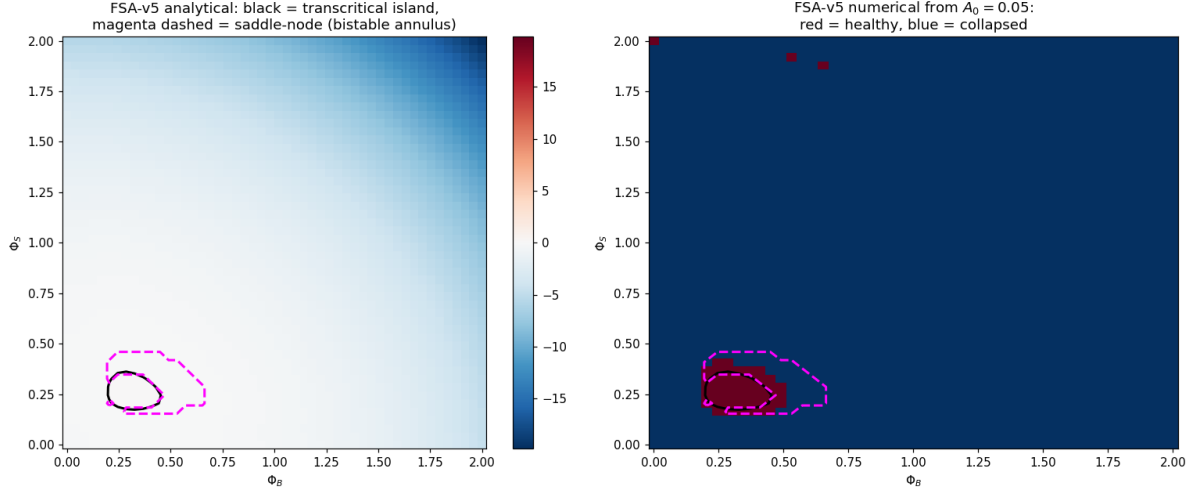


Figure 2: FSA-v5 full bifurcation diagram. Left: analytical transcritical (black) and saddle-node (magenta dashed) contours. Right: numerical regime classification from forward RK4 rollouts starting at $A_0 = 0.05$. Bistable annulus surrounds the healthy island on all sides; both deconditioning and over-training are reachable from a small initial A .

- **Over-training** ($\Phi = (1.0, 1.0)$): F runs away to ≈ 2 in ~ 30 days, $\bar{\mu}$ goes strongly negative via the $-\mu_{FF}(F - F_{\text{Typ}})^2$ term, A collapses by day ~ 50 . Mediated by fatigue, not deconditioning.

4.5 Closed-island sanity check

For verification during a port, the $\bar{\mu}(0; \Phi)$ values at five representative stimuli are tabulated below. These should match exactly when running `TRUTH_PARAMS_V5` through the analytical $\bar{\mu}$ function (e.g., `control_v5.py:_mu_bar`).

Stimulus regime	$\bar{\mu}(0; \Phi)$	Outcome
Sedentary, $\Phi = (0, 0)$	-0.180	strong collapse
Aerobic-only, $\Phi = (0.30, 0)$	-0.059	collapse
Strength-only, $\Phi = (0, 0.30)$	-0.093	collapse
Balanced moderate, $\Phi = (0.30, 0.30)$	+0.011	healthy ($A^* \approx 0.54$)
Over-training, $\Phi = (1.0, 1.0)$	-1.608	catastrophic collapse

5 The probabilistic cost functional

This section is the v5 main novelty. The basin geometry of Section 4 dictates a specific form of cost functional that the `smc2fc` controller should optimise.

5.1 The structurally correct formulation: chance constraints

Given the bistable annulus surrounding the healthy island, the natural clinical question is: *what minimum-effort training programme keeps the athlete in the healthy basin with at least probability $1 - \alpha$?* This is encoded as a chance constraint on basin escape:

$$\mathbb{P}[A_t < A_{\text{sep}}(\Phi_t)] \leq \alpha \quad \forall t \in [0, T]. \quad (21)$$

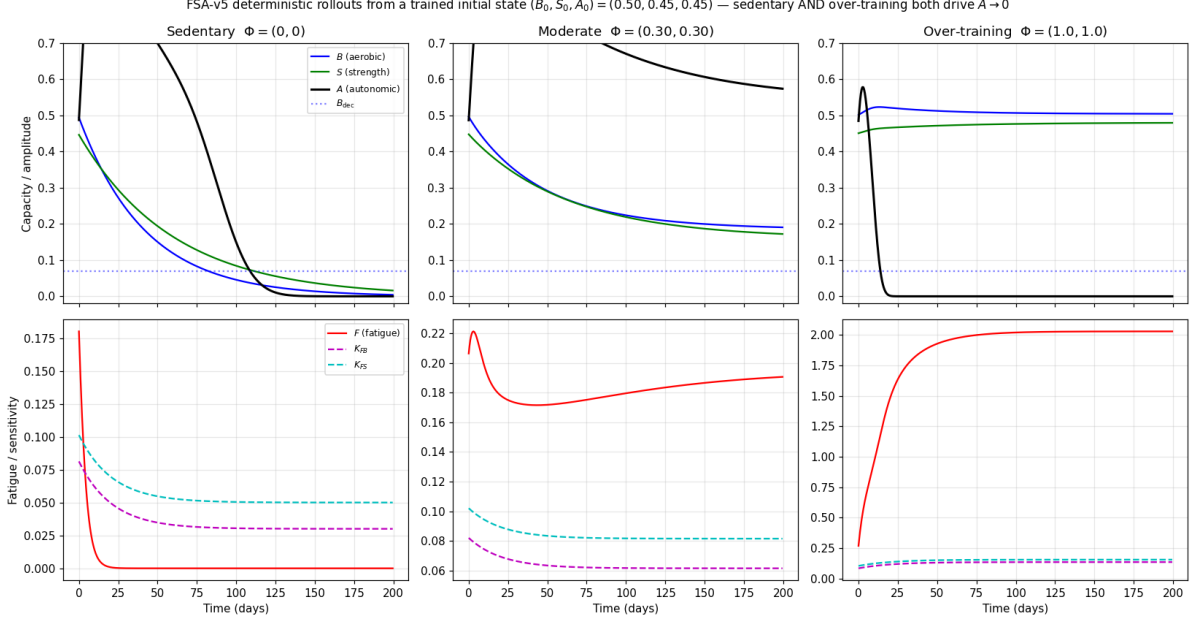


Figure 3: Deterministic RK4 rollouts under the three regimes. Top row: chronic states B, S, A . Bottom row: fatigue F and sensitivities K_{FB}, K_{FS} .

The cost functional pairs naturally with this constraint:

$$\min_{\theta} \lambda_{\Phi} \int_0^T \|\Phi(t)\|^2 dt \quad \text{s.t.} \quad \mathbb{E} \left[\int_0^T A_t dt \right] \geq A_{\text{target}}, \quad \mathbb{P}[A_t < A_{\text{sep}}(\Phi_t)] \leq \alpha. \quad (22)$$

This formulation directly answers the clinical question: minimise total effort subject to (i) achieving at least target autonomic exposure A_{target} in expectation and (ii) keeping the basin-escape probability below the budget α at every time.

5.2 Why this needs SMC², not gradient-OT

The chance constraint (21) is structurally non-differentiable: it involves an expectation over indicator functions of state events. A gradient-based open-loop optimiser cannot represent it; only an SMC² controller, which carries a particle population over both parameters and trajectories, can naturally reject or re-weight θ -particles whose simulated trajectories violate the constraint at the prescribed rate.

5.3 Reference implementations: hard / soft variants

The chance-constraint cost is implemented as *two interchangeable JIT-friendly variants* in `models/fsa_high_res/control_v5.py`, plus a back-compat alias and a legacy NumPy reference. The `smc2fc` controller picks one variant based on its outer-loop architecture.

Hard variant — for pure SMC² architectures. `evaluate_chance_constrained_cost_hard(...)` uses the indicator $\mathbb{I}[A_t < A_{\text{sep}}(\Phi_t)]$ directly. Suitable when the controller’s outer loop weights θ -particles by their empirical violation rate at each tempering step. The hard indicator is non-differentiable in θ , so this variant is NOT for HMC inner kernels. The full implementation is JIT-compiled and `vmap`-ed over particles. Per-bin separatrix root-finding is a JAX bisection over a 64-point grid — no `scipy.optimize.brentq`, no Python loops.

Soft variant — for HMC architectures. `evaluate_chance_constrained_cost_soft(...)` replaces the indicator with a sigmoid surrogate:

$$\mathbb{I}[A_t < A_{\text{sep}}(\Phi_t)] \longrightarrow \sigma\left(\beta \cdot \frac{A_{\text{sep}}(\Phi_t) - A_t}{\text{scale}}\right), \quad (23)$$

which is C^∞ and differentiable through θ via the trajectory A_t . As $\beta \rightarrow \infty$ the surrogate converges to the hard indicator (the smoke test verifies this within 10^{-3} at $\beta = 10^4$). Recommended use: anneal β upward over the inference run — small β early on for smooth easy gradients, large β later for structural correctness. Default $\beta = 50$, $\text{scale} = 0.1$ (units of A).

```

1 def evaluate_chance_constrained_cost_hard(
2     theta_particles,          # list of param dicts, OR dict-of-arrays (
3     n_particles,)            # (n_particles,), normalised to sum to 1
4     Phi_schedule,            # (n_steps, 2)
5     *,
6     dt: float = 1.0 / 96,    # 15-min bins (FSA_STEP_MINUTES = 15)
7     alpha: float = 0.05,     # chance-constraint budget
8     A_target: float = 2.0,   # min E[ integral A dt ]
9     truth_params_template=None, # default TRUTH_PARAMS_V5
10    initial_state=None,       # (6,) - default = trained athlete
11 ) -> dict:
12     # Returns:
13     #   mean_effort          scalar - integral ||Phi||^2 dt
14     #   mean_A_integral      scalar - weighted-mean integral A_t dt
15     #   violation_rate_per_particle (n_particles,)
16     #   weighted_violation_rate scalar
17     #   satisfies_chance_constraint bool - weighted rate <= alpha
18     #   satisfies_target      bool - mean_A_integral >= A_target
19     #   A_sep_per_bin         (n_steps,) - analytical separatrix curve

```

The soft variant has two extra kwargs, `beta` (default 50) and `scale` (default 0.1); the rest of the signature is identical.

Back-compat alias. The original symbol `evaluate_chance_constrained_cost` is preserved as an alias for the hard variant, so existing imports from this module continue to work unchanged. The slow legacy NumPy implementation lives at `_evaluate_chance_constrained_cost_legacy` and is kept only as a debug / readability reference; production callers should use the hard or soft variants.

Architecture choice for the smc2fc port. The decision tree for the smc2fc controller author:

- Outer loop is *pure SMC² with importance weighting* — particles are re-weighted at each tempering step by their per-particle violation rate, no inner gradient kernel. Use **hard** variant.
- Outer loop has *HMC* (or any gradient-based optimiser) inside the SMC tempering — needs $\nabla_\theta J$. Use **soft** variant; anneal β from ~ 10 at the first temperature to $\sim 10^3$ near convergence.
- Need a back-compat baseline for unit tests against the v3 cost functional — use `build_control_spec_v5` (gradient-OT) only as a fallback.

The smc2fc controller is responsible for closing the loop: re-weighting, rejecting, or scoring candidate Φ schedules using the dict returned from either variant. This file does not own the controller; it provides only the cost evaluators.

5.4 Engineering note: deterministic CRN

Both variants forward-simulate each particle deterministically (no per-particle noise) for speed and reproducibility. The chance constraint then captures *parameter uncertainty*, not state-noise uncertainty. To upgrade to full stochastic simulation, replace the internal helper `_make_forward_rollout_fn` with a CRN-noise variant — every other piece of the cost function logic is unchanged.

5.5 Bonus formulation: Lagrangian relaxation

For consumers that need a smooth scalar cost (e.g. for warmstart of the SMC² controller), the chance constraint can be relaxed to a soft penalty:

$$\tilde{J}(\theta) = \lambda_{\Phi} \int \|\Phi\|^2 dt - \mathbb{E} \left[\int A dt \right] + \lambda_{\text{chance}} \max(0, \hat{p}_{\text{violate}} - \alpha)^2, \quad (24)$$

where $\hat{p}_{\text{violate}}$ is the empirical violation rate from the particle cloud and λ_{chance} is a large penalty weight. This loses the chance-constraint guarantee but yields a smooth objective amenable to gradient methods. Equivalent to using the soft variant with $\beta \rightarrow 0$ (very loose surrogate) plus an explicit violation-rate penalty.

6 Numerical integration: time step and stiffness

6.1 Production grid: 15-minute bins

The model is integrated on a 15-minute bin grid:

$$\Delta t = \frac{1}{96} \text{ day} \approx 15 \text{ minutes}, \quad \text{bins per day} = 96. \quad (25)$$

This grid is set by the environment variable `FSA_STEP_MINUTES` (default 15) in `models/fsa_high_res/simulation.py`. The computation is:

```
1 import os as _os
2 _STEP_MIN = int(_os.environ.get('FSA_STEP_MINUTES', '15'))
3 BINS_PER_DAY = (60 * 24) // _STEP_MIN          # = 96 at default
4 DT_BIN_DAYS = 1.0 / BINS_PER_DAY              # = 1/96
```

Each integration step is exactly one bin — there is no inner sub-stepping in the production path (`_plant.py:_plant_em_step`). The choice is justified by stiffness analysis below.

6.2 Why one EM step per bin is safe

The slowest characteristic time is $\tau_S = 60\text{d}$ (strength capacity relaxation). The fastest is $\tau_F = 6.36\text{d}$ (fatigue clearance, after the autonomic correction). The A -equation has a cubic damping term $-\eta A^3$ which can be locally fast for large A , but $A \in [0, \sim 1]$ in practice and $\eta = 0.20$, so the autonomic time constant is no faster than $\sim 5\text{d}$.

The shortest meaningful time-scale is therefore $\sim 6\text{d}$. The integration step is $\Delta t = 1/96 \approx 0.01\text{d}$ — about $600\times$ smaller than the fastest mode. Forward Euler–Maruyama is unconditionally stable on this margin; no implicit or sub-stepped integrator is needed. (For tools that wish to use coarser dt — e.g. the basin sweep tool — a sub-stepped helper is provided as `_dynamics.py:imex_step_substepped`.)

6.3 Why daily sampling is non-viable for inference

Sampling resolution is non-negotiable. Replacing the production 15-minute observation grid with daily summaries inflates the FIM condition number from $\sim 10^{18}$ to $\sim 10^{38}$ and creates an artificial (ϵ_{AS}, μ_S) confound that the production sampling does not have. Two specific reasons:

1. Dense overnight resting-HR sampling pins down aerobic fitness B from passive autonomic signals — a much more informative window than sparse daytime spot-checks.
2. Time-resolved sleep-gated HR separates the (ϵ_{AS}, μ_S) contributions to autonomic amplitude that are otherwise confounded.

Recommendation for the smc2fc port: keep `FSA_STEP_MINUTES = 15` and accept the higher computational cost (8064 SDE steps over 84 days versus ~ 340 at $\Delta t = 0.25\text{d}$).

6.4 State clipping after each step

The post-Euler–Maruyama state is clipped back into the physical domain at each bin:

- $B \in [\epsilon_B, 1 - \epsilon_B]$ (Jacobi reflection or clip)
- $S \in [\epsilon_S, 1 - \epsilon_S]$ (same)
- $F, A, K_{FB}, K_{FS} \in [0, \infty)$ ($\max(\cdot, 0)$)

with $\epsilon_B = \epsilon_S = \epsilon_A = 10^{-4}$. The reflection on B, S matches the natural behaviour of the Jacobi diffusion process at boundaries; the half-line clipping for F, A, K is a safety net since the CIR diffusion is degenerate at zero.

6.5 Numerical convention summary

Quantity	Value (production)
Time unit	days
Bin width Δt	1/96 d ≈ 15 min
Bins per day	96
Sub-stepping in plant	none (single EM step per bin)
Sub-stepping in tools	4 sub-steps optional via <code>imex_step_substepped</code>
Reflection: B, S	$\epsilon = 10^{-4}$ on both ends
Floor: F, A, K_*	0, with A getting a $+\epsilon_A$ floor in diffusion only
Random number generator	JAX (CRN-friendly via <code>jax.random.PRNGKey</code>)
Float precision	enable via <code>JAX_ENABLE_X64=True</code> (Section 9)

7 Parameter set: 37 estimated, 14 frozen

This section explains the pinning strategy used in `HIGH_RES_FSA_V5_ESTIMATION`. Read the appendix for the full symbol-by-symbol table; this section gives the rationale.

7.1 The split

Estimated (37)	in <code>PARAM_PRIOR_CONFIG</code>
Frozen (14)	in <code>frozen_params</code>

The 14 frozen parameters fall into two groups:

1. **Diffusion-side (6 always frozen):** $\sigma_B, \sigma_S, \sigma_F, \sigma_A, \sigma_K, \phi$. Diffusion identifiability is weak under SMC² in any setting and is not the right thing to learn jointly with drift.
2. **Dynamics-side (8 frozen by FIM analysis + first-port-test policy):** $K_{FB}^0, K_{FS}^0, \tau_K, n_{\text{dec}}, B_{\text{dec}}, S_{\text{dec}}$

7.2 Why each dynamics-side freeze

K_{FB}^0 and K_{FS}^0 (**structurally non-identifiable**). The K subsystem rapidly relaxes onto its slow manifold $K_{Fi}^* = K_{Fi}^0 + \tau_K \mu_K \Phi_i$. The data observes K_{Fi} only indirectly through the F -equation, and only ever sees the *combination* $K_{Fi}^0 + \tau_K \mu_K \Phi_i$ — never K_{Fi}^0 in isolation. No amount of additional data fixes this: the FIM eigenvector for these is structurally zero. Pin them to physiologically sensible values ($K_{FB}^0 = 0.030$, $K_{FS}^0 = 0.050$) and forget about estimating them.

τ_K (**jointly weakly identified with μ_K**). At steady state, only the product $\tau_K \mu_K$ is observable from K relaxation. To separate them you need calibration data with sustained loading periods of order $\tau_K \approx 21$ days, which is rarely available. Pin τ_K to the Busso-standard 21d and estimate only μ_K .

n_{dec} (**structural shape**). The Hill exponent is a structural shape parameter, not a data-fitted one. The transition steepness it controls is a modelling choice. Pin at $n_{\text{dec}} = 4$.

$B_{\text{dec}}, S_{\text{dec}}, \mu_{B-}, \mu_{S-}$ (**need long detraining episodes**). The four v5 deconditioning parameters are only informable from data that includes long (≥ 6 weeks) detraining episodes, because the Hill term in equation (7) is silent above the threshold. For a first port-test on simple steady-state synthetic data, pin them all at the §10 Figure 7 calibration values: $(B_{\text{dec}}, S_{\text{dec}}, \mu_{B-}, \mu_{S-}) = (0.07, 0.07, 0.10, 0.10)$.

7.3 Relaxation policy: how to widen the parameter set later

When richer calibration data becomes available, freezes can be relaxed one at a time by moving a key from `frozen_params` into `PARAM_PRIOR_CONFIG`. Suggested priorities, in order of unblockability:

1. μ_{B-}, μ_{S-} if a long detraining episode exists in the data.
2. $B_{\text{dec}}, S_{\text{dec}}$ if the detraining episode crosses the threshold smoothly.
3. τ_K if a ≥ 21 d sustained-load episode is available.
4. K_{FB}^0, K_{FS}^0 — never. Structurally non-identifiable.

7.4 Quick reference: what's in the parameter vector

The `HIGH_RES_FSA_V5_ESTIMATION` parameter vector is 37 long. Component-wise view:

Group	Count	Members (briefly)
Aerobic-capacity dynamics	3	$\tau_B, \kappa_B, \epsilon_{AB}$
Strength-capacity dynamics	3	$\tau_S, \kappa_S, \epsilon_{AS}$
Fatigue dynamics	2	τ_F, λ_A
Variable-dose K dynamics	1	μ_K (τ_K frozen, K_{Fi}^0 frozen)
Stuart–Landau drive	6	$\mu_0, \mu_B, \mu_S, \mu_F, \mu_{FF}, \eta$
HR observation	5	$\text{HR}_{\text{base}}, \kappa_B^{\text{HR}}, \alpha_A^{\text{HR}}, \beta_C^{\text{HR}}, \sigma_{\text{HR}}$
Sleep Bernoulli	3	$k_C, k_A, \tilde{c}$
Stress observation	5	$S_{\text{base}}, k_F, k_{A,S}, \beta_C^S, \sigma_S$
Steps observation	6	$\mu_{\text{step},0}, \beta_B^{\text{st}}, \beta_F^{\text{st}}, \beta_A^{\text{st}}, \beta_C^{\text{st}}, \sigma_{\text{st}}$
Volume Load observation	3	$\beta_S^{\text{VL}}, \beta_F^{\text{VL}}, \sigma_{\text{VL}}$
Total estimated	37	

8 Test scenarios for synthetic data

This section gives concrete test scenarios that exercise the v5 implementation end-to-end. Each one is a constant- Φ rollout from a defined initial state, with a predicted qualitative outcome that the engineer can verify against. The test infrastructure is in `tests/test_fsa_v5_smoke.py`.

8.1 Test 1: Sedentary collapse

Setup: Trained-athlete initial state $(B, S, F, A, K_{FB}, K_{FS}) = (0.50, 0.45, 0.20, 0.45, 0.06, 0.07)$. Constant $\Phi = (0, 0)$. Run for 200 days.

Expected: B and S decay exponentially at $\tau_B = 42\text{d}$ and $\tau_S = 60\text{d}$. Around day 60–100, both fall below the deconditioning threshold $B_{\text{dec}} = S_{\text{dec}} = 0.07$ and the Hill terms in μ saturate. With $\bar{\mu}$ now strongly negative, A collapses to near zero by day ~ 150 . Fatigue F also decays (no input). K states relax back to baseline.

Pass criterion: $A(t = 200\text{d}) < 0.01$, but $A(t = 30\text{d}) > 0.3$ (not collapsed yet) — demonstrates the slow-detraining time-course.

Code:

```
1 from models.fsa_high_res import StepwisePlant, DEFAULT_PARAMS_V5
2 import numpy as np
3
4 plant = StepwisePlant(truth_params=dict(DEFAULT_PARAMS_V5))
5 plant.state = np.array([0.50, 0.45, 0.20, 0.45, 0.06, 0.07])
6 out = plant.advance(stride_bins=200 * 96,
7                      Phi_daily=np.zeros((200, 2)))
8 A_traj = out['trajectory'][:, 3]
9 assert A_traj[-1] < 0.01, f"A did not collapse: {A_traj[-1]}"
10 assert A_traj[30 * 96] > 0.3, f"A collapsed too early"
```

8.2 Test 2: Moderate (healthy) attractor

Setup: Same trained-athlete initial state. Constant $\Phi = (0.30, 0.30)$ (inside the v5 healthy island). Run for 100 days.

Expected: B and S relax to slow-manifold values near 0.18 each. $\bar{\mu}$ stays positive ($\sim +0.011$). A relaxes to its healthy attractor $A^* \approx 0.55$ over the first ~ 50 days. F stabilises near $F_{\text{TYP}} = 0.20$. K states rise modestly.

Pass criterion: $A(100\text{d}) > 0.4$, $|F(100\text{d}) - 0.20| < 0.10$, no NaN.

8.3 Test 3: Over-training collapse

Setup: Same trained-athlete initial state. Constant $\Phi = (1.0, 1.0)$ (well outside the v5 island). Run for 60 days.

Expected: F runs away to ≈ 2 within 30 days. The $-\mu_{FF}(F - F_{\text{TYP}})^2$ term in μ dominates, $\bar{\mu}$ goes strongly negative (~ -1.6), and A collapses to zero by day ~ 50 . B, S saturate at their upper bounds.

Pass criterion: $A(60\text{d}) < 0.01$, but $F(60\text{d}) > 1.5$ (the collapse mechanism is fatigue-driven, not deconditioning).

8.4 Test 4: Bistability sensitivity (separatrix crossing)

Setup: Constant Φ chosen inside the bistable annulus (e.g. $\Phi = (0.50, 0.30)$ from §7 of `main.tex`). Run from two different initial states differing only in A_0 :

- Trial A: $A_0 = 0.10$ (above separatrix)
- Trial B: $A_0 = 0.02$ (below separatrix)

Expected: Trial A converges to the healthy interior attractor; Trial B collapses to $A = 0$. Same training, different outcomes — this is the bistability test.

Pass criterion: $|A_{\text{trialA}}(150\text{d}) - 0.5| < 0.2$ and $A_{\text{trialB}}(150\text{d}) < 0.01$.

8.5 Test 5: Closed-loop with the chance-constrained cost

Setup: Build a 10-particle cloud of parameters all at the v5 truth (no inference yet). Build a 14-day moderate $\Phi = (0.30, 0.30)$ schedule. Call `evaluate_chance_constrained_cost` with $\alpha = 0.05$, $A_{\text{target}} = 2.0$.

Expected: `satisfies_chance_constraint = True`, `satisfies_target = True`, `weighted_violation_` ≈ 0 , `mean_A_integral` ≈ 11 ($\approx A^* \cdot 14\text{d} \cdot 96 \text{ bins} \cdot \text{dt}$).

This test exercises the entire v5 cost-evaluation path end-to-end.

8.6 Recommended order to run during the port

1. Test 2 (moderate, healthy) — confirms the implementation can produce a stable trajectory.
2. Test 1 (sedentary collapse) — confirms the Hill term fires correctly under low Φ .
3. Test 3 (over-training) — confirms the v4-inherited fatigue mechanism is intact.
4. Test 4 (bistability) — confirms the closed-island geometry actually exists at runtime.
5. Test 5 (cost evaluator) — confirms the cost evaluator wires up correctly.

If any of these fails, the port has an issue; consult Section 9.

9 Common bugs and gotchas

This section documents specific issues encountered during the v5 implementation pass. Read it before you port; each item below is a real bug that took non-trivial effort to surface.

9.1 The `sigma_S` name collision

Symptom: The strength state S in the latent SDE develops 500× larger noise than expected; trajectory shows wild swings of order ± 0.5 over a single day.

Cause: The `DEFAULT_PARAMS` dictionary in `simulation.py` contains *two* keys named `sigma_S` — one for the latent-S Jacobi diffusion scale (~ 0.008) and another for the stress-channel observation noise (~ 4.0). Python’s dict keeps only the last assignment, which is the observation one. Reading `truth_params['sigma_S']` therefore returns 4.0 (the obs noise) where the SDE integrator expects the diffusion scale.

Fix: The plant in `_plant.py` hard-codes the diffusion scales as constants matching `SIGMA*_FROZEN` in `estimation.py`, bypassing the params dict entirely. The `estimation.py` `propagate_fn` also hard-codes them. If you write new code that needs the diffusion scales, follow this convention — do not read them from the params dict.

9.2 `JAX_ENABLE_X64` required

Symptom: “Explicitly requested dtype float64 ... is not available, and will be truncated to dtype float32” warnings spam the console; numerical results subtly disagree with reference values.

Cause: JAX defaults to float32 unless either the environment variable `JAX_ENABLE_X64=True` is set or `jax.config.update("jax_enable_x64", True)` is called before JAX is imported.

Fix: Set the environment variable, or do


```

1 import os
2 os.environ['JAX_ENABLE_X64'] = 'True'
3 from jax import config
4 config.update('jax_enable_x64', True)

```

at the top of any tool/test that runs the model. The smoke test (`tests/test_fsa_v5_smoke.py`) does this.

9.3 EstimationModel requires init_state_prior_config

Symptom: `TypeError: EstimationModel.__init__() missing 1 required positional argument: 'init_state_prior_config'.`

Cause: The `EstimationModel` dataclass in `smc2fc/estimation_model.py` has `init_state_prior_config` as a required (not optional) constructor argument.

Fix: Pass an empty `OrderedDict()` for it if no init-state prior is desired. The v5 `HIGH_RES_FSA_V5_ESTIMATION` in `estimation.py` does exactly this.

9.4 Drift duplication — single source of truth

Symptom: Subtle drift differences between the JAX path and the numpy path that only manifest at certain parameter regimes.

Cause: The drift function appears in three places: `_dynamics.py:drift_jax` (canonical), `simulation.py:drift` (numpy mirror), and `simulation.py:drift_jax` (JAX wrapper for `SDEModel` API). All three must be kept in sync. The v5 implementation delegates from `estimation.py` to the canonical `_dynamics.drift_jax` via factory pattern, but the `simulation.py` copies are still inlined (because they have a different calling convention).

Fix: If you modify the drift, modify all three places. There is a comment block at the top of `simulation.py:drift` that says “KEEP IN SYNC with `_dynamics.drift_jax`”. Read that. The chance-constraint cost in `control_v5.py` also re-derives $\bar{\mu}$ in a fourth place — keep that one in sync too.

9.5 Plant truth_params must include observation coefficients

Symptom: `KeyError: 'k_C'` (or any obs coefficient) when running `StepwisePlant.advance()`.

Cause: The plant calls the obs samplers (`gen_obs_sleep`, `gen_obs_hr`, ...), which read observation coefficients from `self.truth_params`. If you initialise the plant with `truth_params=TRUTH_PARAMS` (from `_dynamics.py`), it has only the dynamics + diffusion keys — not the obs coefficients.

Fix: Initialise the plant with `truth_params=DEFAULT_PARAMS_V5` (from `simulation.py`), which has both. Or merge: `dict(**DEFAULT_PARAMS_V5, **TRUTH_PARAMS_V5)`.

9.6 K_{FB}^0 and K_{FS}^0 are structurally non-identifiable

Symptom: The posterior on K_{FB}^0 and K_{FS}^0 looks identical to the prior; condition number of the FIM blows up.

Cause: As Section 7 explains, the data only ever sees the combination $K_{Fi}^0 + \tau_K \mu_K \Phi_i$. The constants K_{Fi}^0 are formally non-identifiable.

Fix: Pin them. The default v5 `frozen_params` dict already does this (see `estimation.py`). Do not be tempted to add them to `PARAM_PRIOR_CONFIG` no matter how rich the data is.

9.7 Don’t collapse observations to daily summaries during inference

Symptom: Suspiciously bad posterior identifiability; (ϵ_{AS}, μ_S) pair always shows up in the FIM smallest eigenvectors.

Cause: Switching from the production 15-minute bin grid to daily summary observations destroys the dense overnight HR signal that pins B and the time-resolved sleep-gated HR that separates ϵ_{AS} from μ_S (Section 6).

Fix: Keep `FSA_STEP_MINUTES = 15`. Pay the $\sim 24\times$ extra computational cost. The FIM analysis (§11 of `main.tex`) shows this changes the condition number by 20 orders of magnitude.

9.8 Particle parameter dicts must include v5 Hill keys

Symptom: `KeyError: 'B_dec'` (or any v5 Hill key) when forward-simulating particles.

Cause: The canonical `drift_jax` in `_dynamics.py` expects all v5 Hill keys present (`B_dec`, `S_dec`, `mu_dec_B`, `mu_dec_S`, `n_dec`). If the SMC² engine carries θ -particles whose dicts only contain the estimated parameters (37 of them, Section 7), the four v5 Hill keys are missing and the drift call fails.

Fix: Always merge with the frozen-params dict before calling `drift_jax`. The cost evaluator `evaluate_chance_constrained_cost` in `control_v5.py` already does this; new consumers need to follow suit. Pattern:

```
1 p_full = {**FROZEN_DICT, **estimated_dict}
2 y_next = y + dt * drift_jax(y, p_full, Phi_t)
```

9.9 Observation gating must match production

Symptom: FIM analysis under-estimates information content; inference posterior is wider than expected.

Cause: The four Gaussian observation channels are gated by sleep/wake state (HR sleep-only, Stress and Steps wake-only, Volume Load training-session-only). If your inference pipeline ingests the observations as if they were emitted at every bin, the Kalman fusion weights them all and the FIM looks fatter than it really is. The opposite mistake (treating them as daily averages) destroys identifiability as in §6.

Fix: Use the `align_obs_fn` in `estimation.py:align_obs_fn` to pack scattered observations into bin-resolved arrays with explicit `*_present` masks. The `propagate_fn_v5` respects the masks correctly.

9.10 Initial state defaults: bounded but “deconditioned”

The default initial state in `DEFAULT_INIT` ($B = 0.05, S = 0.10, F = 0.30, A = 0.10, K_{FB} = 0.030, K_{FS} = 0.050$) is a “deconditioned but otherwise healthy” starting point, not a trained athlete. Do not be surprised if a forward sim from `DEFAULT_INIT` under moderate Φ takes weeks to settle — the system is climbing from low capacity. For testing closed-loop behaviour, override the initial state with a trained-athlete vector (Section 8).

Appendix: Complete FSA-v5 Parameter Reference

This appendix is the canonical reference for every parameter in the FSA-v5 model as implemented in the “smc2fc“-port-ready codebase. For each parameter we list its symbol, role in the equations of motion, prior (if estimated) or frozen value, and where it first appears in the main document.

The parameter set as implemented in `models/fsa_high_res/estimation.py` consists of **37 estimated parameters** plus **14 frozen** (8 dynamics + 6 diffusion-related). The maximum-pinning rationale is the §11.6 recommendation derived from the Fisher-information analysis in §11.

Estimated parameters (37)

These are the entries of `PARAM_PRIOR_CONFIG` in `estimation.py`. Lognormal priors are parameterised as $\log \theta \sim \mathcal{N}(\mu_{\log}, \sigma_{\log}^2)$ — the table shows $(\exp(\mu_{\log}), \sigma_{\log})$. Normal priors are parameterised directly as (μ, σ) .

Symbol	Role / location	Prior	Truth
<i>Dynamics — aerobic capacity B</i>			
τ_B	B time-scale / §11.1	lognormal (42.0, 0.10)	42.0 d
κ_B	B gain / §11.1	lognormal (0.01248, 0.20)	0.01248
ϵ_{AB}	autonomic-coupling on B / §11.1	lognormal (0.40, 0.05)	0.40
<i>Dynamics — strength capacity S</i>			
τ_S	S time-scale / §11.1	lognormal (60.0, 0.10)	60.0 d
κ_S	S gain / §11.1	lognormal (0.00816, 0.20)	0.00816
ϵ_{AS}	autonomic-coupling on S / §11.1	lognormal (0.20, 0.05)	0.20
<i>Dynamics — fatigue F</i>			
τ_F	F time-scale / §11.1	lognormal (6.36, 0.15)	6.36 d
λ_A	autonomic-coupling on F / §11.1	lognormal (1.00, 0.05)	1.00
<i>Dynamics — variable-dose K (Busso 2003)</i>			
μ_K	K damage rate / §11.1	lognormal (0.005, 0.20)	0.005
<i>Dynamics — Stuart–Landau bifurcation parameter $\bar{\mu}$</i>			
μ_0	baseline drive / §11.1	lognormal (0.036, 0.20)	0.036
μ_B	B→A coupling / §11.1	lognormal (0.30, 0.20)	0.30
μ_S	S→A coupling / §11.1	lognormal (0.15, 0.20)	0.15
μ_F	F→A linear suppression / §11.1	lognormal (0.26, 0.20)	0.26
μ_{FF}	quadratic F penalty / §11.1	lognormal (0.40, 0.05)	0.40
η	cubic damping in A / §11.1	lognormal (0.20, 0.15)	0.20
<i>HR observation channel</i>			
HR_{base}	resting HR baseline (bpm) / §11.1	normal (62.0, 2.0)	62.0
κ_B^{HR}	HR $\propto -B$ slope / §11.1	lognormal (12.0, 0.15)	12.0
α_A^{HR}	HR $\propto +A$ slope / §11.1	lognormal (3.0, 0.20)	3.0
β_C^{HR}	HR circadian coefficient / §11.1	normal (−2.5, 0.5)	−2.5
σ_{HR}	HR observation noise (bpm) / §11.1	lognormal (2.0, 0.20)	2.0
<i>Sleep Bernoulli channel</i>			
k_C	sleep $\propto C$ slope / §11.1	lognormal (3.0, 0.15)	3.0
k_A	sleep $\propto A$ slope / §11.1	lognormal (2.0, 0.25)	2.0
$\tilde{c}$	sleep logit intercept / §11.1	normal (0.5, 0.25)	0.5
<i>Stress observation channel</i>			
S_{base}	stress baseline / §11.1	normal (30.0, 3.0)	30.0
k_F	stress $\propto +F$ slope / §11.1	lognormal (20.0, 0.20)	20.0
$k_{A,S}$	stress $\propto -A$ slope / §11.1	lognormal (8.0, 0.25)	8.0
β_C^S	stress circadian coefficient / §11.1	normal (−4.0, 0.8)	−4.0
σ_S	stress observation noise / §11.1	lognormal (4.0, 0.20)	4.0
<i>Steps observation channel (log-Gaussian)</i>			
$\mu_{\text{step},0}$	log-steps intercept / §11.1	normal (5.5, 0.3)	5.5
β_B^{st}	steps $\propto +B$ slope / §11.1	lognormal (0.8, 0.20)	0.8
β_F^{st}	steps $\propto -F$ slope / §11.1	lognormal (0.5, 0.25)	0.5
β_A^{st}	steps $\propto +A$ slope / §11.1	lognormal (0.3, 0.25)	0.3
β_C^{st}	steps circadian coefficient / §11.1	normal (−0.8, 0.2)	−0.8
σ_{st}	log-steps noise / §11.1	lognormal (0.5, 0.15)	0.5
<i>Volume Load observation channel</i>			
β_S^{VL}	VL $\propto +S$ slope / §11.1	lognormal (100.0, 0.15)	100.0
β_F^{VL}	VL $\propto -F$ slope / §11.1	lognormal (20.0, 0.20)	20.0
σ_{VL}	VL observation noise / §11.1	lognormal (10.0, 0.20)	10.0

Frozen parameters (8 dynamics + 6 diffusion-related)

These do not appear in the estimated parameter vector. They are baked into the “Estimation-Model.frozen_params” dict. Reasons for freezing each one are tied directly to the §11 FIM analysis.

Symbol	Frozen value	Role / location	Reason for freezing
<i>Dynamics-side freezes</i>			
K_{FB}^0	0.030	baseline aerobic K / §11.1	Structurally non-identifiable. Slow-manifold relaxation collapses K_{FB}^0 and $\tau_K \mu_K$ into a single observable. §11.6 result (1).
K_{FS}^0	0.050	baseline strength K / §11.1	Same as K_{FB}^0 .
τ_K	21.0 d	K relaxation timescale / §11.1	Busso-standard. Jointly weakly identified with μ_K (§11.6 result 5).
n_{dec}	4.0	Hill exponent / §10 eq. 25	Structural shape parameter; not learned.
B_{dec}	0.07	B deconditioning threshold / §10	Pinned to §10.4 Table 5 value. Inference requires ≥ 6 -week detraining episode (§11.6 result 3).
S_{dec}	0.07	S deconditioning threshold / §10	Same.
μ_{B-}	0.10	B deconditioning amplitude / §10	Pinned to §10.4 Table 5 value. Inference requires same long detraining (§11.6 result 2).
μ_{S-}	0.10	S deconditioning amplitude / §10	Same.
<i>Diffusion-side freezes (always frozen, both v4 and v5)</i>			
σ_B	0.010	B Jacobi diffusion / §11.1	Identifiability via diffusion is weak; standard practice.
σ_S	0.008	S Jacobi diffusion / §11.1	Same.
σ_F	0.012	F CIR diffusion / §11.1	Same.
σ_A	0.020	A CIR diffusion / §11.1	Same.
σ_K	0.005	K CIR diffusion (shared) / §11.1	Same.
ϕ	0.0	circadian phase reference / §11.1	Held at zero by convention.

Initial state

The 6D initial state is set from data or from DEFAULT_INIT; it has no explicit prior in the v5 EstimationModel (i.e., INIT_STATE_PRIOR_CONFIG = OrderedDict()). Default values correspond to a “deconditioned but not unhealthy” baseline:

State	Default	Bounds
B	0.05	$[0, 1]$
S	0.10	$[0, 1]$
F	0.30	$[0, \infty)$
A	0.10	$[0, \infty)$
K_{FB}	0.030	$[0, \infty)$
K_{FS}	0.050	$[0, \infty)$

Reading guide for the smc2fc port

After importing models.fsa_high_res, the port has access to:

- HIGH_RES_FSA_V5_ESTIMATION — the “EstimationModel” to register with the SMC² engine. Built from the parameter prior config and frozen-params dict above.

- `HIGH_RES_FSA_V5_MODEL` — the “SDEModel“ for forward simulation.
- `StepwisePlant` — the closed-loop plant for synthetic-feedback testing.
- `evaluate_chance_constrained_cost` — the \$9.6 v5 control cost (equations 23, 24).

The frozen values can be relaxed one at a time (move from `frozen_params` into `PARAM_PRIOR_CONFIG`) as the calibration data set grows to support them.